

◆ Arithmetic Logic Questions

1. Write a Python program to input two numbers and display their **sum, difference, product, and division**.
 2. Write a program to input a number and find its **square and cube**.
 3. Write a program to input the length and breadth of a rectangle and calculate its **area and perimeter**.
 4. Write a program to input the radius of a circle and calculate its **area and circumference**. Use 3.14 for π .
 5. Write a program to input the price of an item and quantity purchased. Calculate the **total bill amount**.
 6. Write a program to input a number of seconds and convert it into **minutes and remaining seconds**.
 7. Write a program to input a three-digit number and calculate the **sum of its digits** using arithmetic operators.
 8. Write a program to input a two-digit number and **reverse the number** using % and //.
 9. A student has marks in 5 subjects. Write a program to calculate the **total marks and percentage**.
 10. Write a program to input the total number of days and calculate approximately how many **weeks and remaining days** it contains.
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◆ Logical Operator Questions

11. Write a program to input a person's age and check whether they are **between 18 and 60 years old** using and.
 12. Write a program to check whether a number is **less than 10 OR greater than 50** using or.
 13. Write a program to check whether a number is **not equal to 0** using the not operator.
 14. Write a program to input marks and check whether a student has passed if **marks are greater than or equal to 40 AND attendance is greater than or equal to 75%**.
 15. Write a program to check whether a number is **divisible by both 3 and 5**.
 16. Write a program to check whether a person can enter a movie if their **age is 18 or above OR they are accompanied by a parent**.
 17. Write a program to check whether a number lies between **1 and 100**.
 18. Write a program to check whether a number is **positive and even**.
 19. Write a program to check whether a student qualifies for admission if:
 - percentage is at least 60
 - AND entrance score is at least 50.
 20. Write a program to check whether a password is valid if its length is at least 8 characters **AND** it is not equal to "password".
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◆ Basic Syntax & Statement Logic

21. Write a program that takes the user's **name and age** and displays:

Hello Karan, you are 20 years old.

22. Write a program to input two numbers and exchange their values **without using a third variable**.

23. Find the output:

```
a = 10
```

```
b = 3
```

```
print(a + b * 2)
```

24. Find the output:

```
x = 20
```

```
y = 5
```

```
x = x + y
```

```
y = x - y
```

```
print(x, y)
```

25. Find the output:

```
a = 15
```

```
b = 4
```

```
print(a // b)
```

```
print(a % b)
```

```
print(a ** 2)
```

26. Find the error and correct the code:

```
name = "Karan"
```

```
age = 20
```

```
print("My name is " name)
```

```
print("My age is", age)
```

27. Write a program to input a number and display whether it is **greater than 100, equal to 100, or less than 100**.

28. Write a program to calculate the final price of an item after applying a **10% discount**.

29. A shop gives a ₹500 discount if the purchase amount is ₹5000 or more. Write a program to calculate the **final amount to pay**.

30. Write a program that takes a student's marks in three subjects and checks whether the student has passed **all three subjects**, where the minimum passing mark is 40 in each subject.

🔥 Extra Challenge Questions

31. Take a four-digit number and calculate the **first digit and last digit**, then find their sum.

32. Input a number and check whether it is **divisible by 2 OR divisible by 7**.

33. Input the cost price and selling price of an item. Calculate the **profit or loss amount**.
34. Input basic salary and calculate:
- HRA = 20% of salary
 - DA = 10% of salary
 - Gross Salary = Basic + HRA + DA
35. Input a person's age and salary. Check whether they are eligible for a loan if **age is between 21 and 60 AND salary is ₹25,000 or more**.
36. Input three numbers and determine whether **all three numbers are equal** using logical operators.
37. Input a number and check whether it is a **two-digit number**.
38. Input the total bill and amount paid. Calculate the **remaining amount to be paid**.
39. Input temperature in Celsius and convert it into Fahrenheit using:
- $$F = (C \times 9/5) + 32$$
40. Input a number and check whether it is **even, positive, and less than 100** using a combination of logical operators.